SOFTWARE ENGINEERING – LAB 3

Architecture Selection: We chose Layered Architecture for the Blood Bank Inventory & Emergency Donor Matcher System.

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Architectural Choice and Reasons

1. Separation of System Functions: Layered architecture divides the Blood Bank system into presentation, request-processing, service-management, and data-access responsibilities. Each layer has a focused role, so changes to one part of the system can be made with less impact on the other parts.

2. Better Control of Emergency Operations: The architecture allows emergency requests to pass through a defined processing layer before inventory and donor services are accessed. This makes important operations such as stock verification, blood allocation, donor matching, and notification easier to coordinate and manage.

Security Advantage

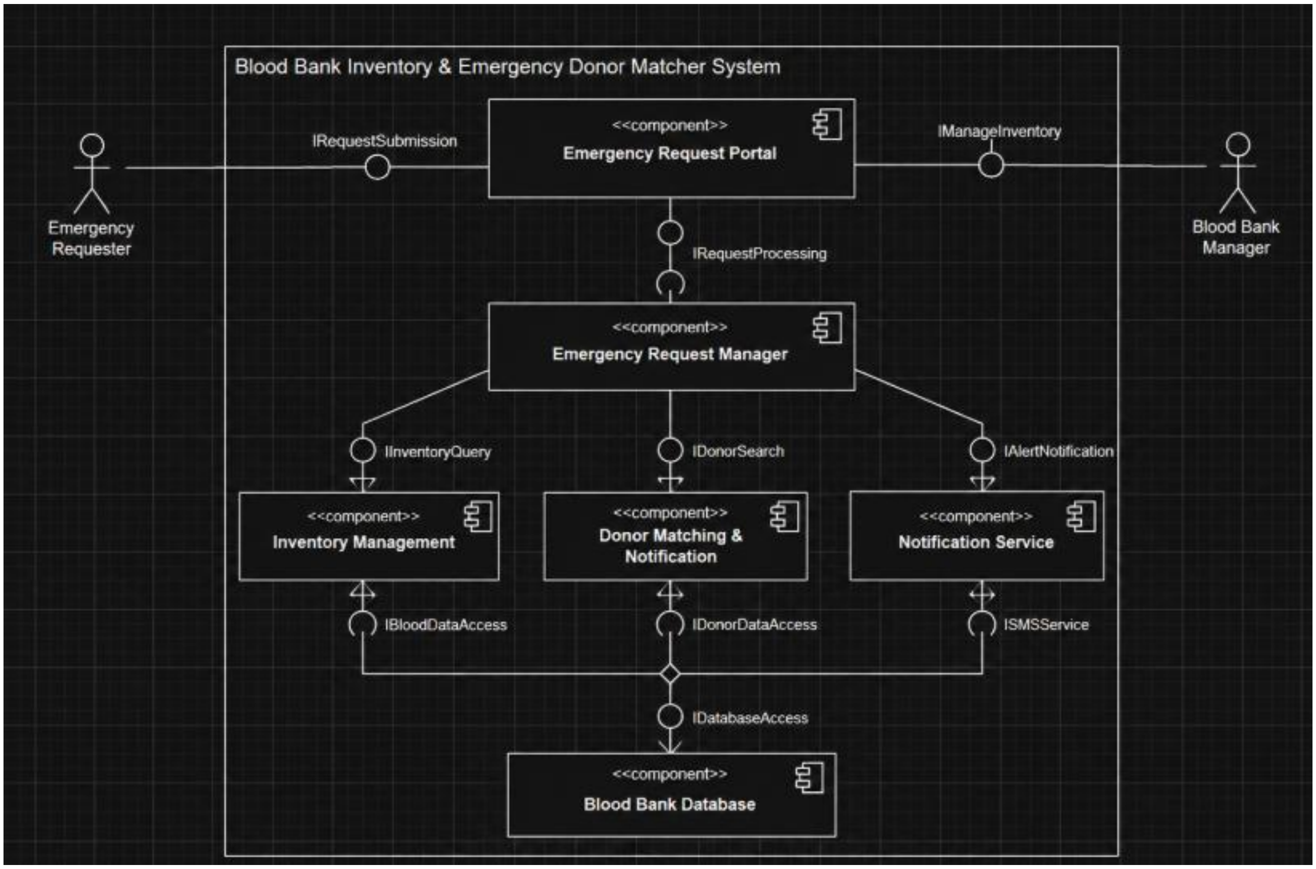
The layered structure provides controlled access between the user-facing part of the system and the blood-bank data. Donor details and inventory records can be accessed only through authorized service components, reducing direct exposure of sensitive information and making access control easier to enforce.

Performance Benefit

Frequently used operations such as checking blood availability can be handled by dedicated service components and optimized database queries. This reduces unnecessary communication between unrelated parts of the system and helps the system respond quickly when an emergency request needs immediate processing.

Component Diagram

The component diagram represents the Blood Bank Inventory & Emergency Donor Matcher System using six main components: Emergency Request Portal, Emergency Request Manager, Inventory Management, Donor Matching & Notification, Notification Service, and Blood Bank Database. It shows the provided and required interfaces between these components and the flow of emergency-request, inventory, donor, notification, and database information.



Component Diagram Description

Emergency Request Portal Component: Provides the user-facing entry point for submitting emergency blood requests and accessing blood availability information.

Emergency Request Manager Component: Processes emergency requests, coordinates inventory checks and blood allocation, and invokes donor-matching services when the required stock is unavailable.

Inventory Management Component: Maintains blood-unit availability, allocation records, and component shelf-life information so that current inventory status can be provided to the request manager.

Donor Matching & Notification Component: Searches for eligible and compatible donors based on the emergency requirement and location, and coordinates the preparation of donor alerts.

Notification Service Component: Delivers emergency notifications to matched donors through the supported SMS notification service.

Blood Bank Database Component: Provides persistent storage and retrieval for emergency requests, blood inventory, donor information, allocation records, and related system data.